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Effect of Feature Extraction on Underwater Moving Body Cavitation Pressure Reconstruction and Prediction

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Abstract: The analysis of cavitation pressure for high-speed underwater moving bodies is critical for hydrodynamic performance and stability. However, traditional methods relying on physical experiments or high-fidelity CFD simulations are computationally expensive and time-consuming, often resulting in small datasets that challenge data-driven approaches. This study aimed to evaluate efficient feature extraction techniques to overcome the limitations of small-sample scenarios. This paper systematically evaluated three feature extraction methods—Principal Component Analysis (PCA), Fast Independent Component Analysis (Fast ICA), and a customized 1-dimensional Convolutional Auto-encoder (Conv1D AE)—for processing cavitation pressure data. The evaluation was conducted through two experiments: the first assessed the ability of these methods to reconstruct the pressure evolution across the body surface in an unsupervised manner, while the second investigated their performance in predicting the peak pressure using a small set of labeled samples. The findings demonstrated a clear trade-off: Fast ICA exhibited the best performance in reconstructing the overall pressure evolution, followed closely by PCA, while the Conv1D AE showed limitations in capturing sharp pressure gradients. Conversely, for the critical task of peak pressure prediction from limited labeled data, the Conv1D AE model achieved a significant 10% increase in accuracy compared to the baseline model without feature extraction, with PCA providing a 3% improvement. Fast ICA, however, was less effective for this specific prediction task. These results underscore the effectiveness of tailored feature extraction in automating cavitation analysis. By reducing reliance on manual intervention and accelerating the extraction of key features like peak pressure, these methods offer a practical pathway to enhance the design and analysis cycle of underwater moving bodies. The study establishes a foundation for applying machine learning to small-sample fluid mechanics problems, with future work focused on optimizing network architectures for improved precision.

Keywords: Conv1D Auto-encoder; Small sample; Cavitation, Fast ICA, PCA

1. Introduction

Cavitation, a complex phenomenon in fluid mechanics characterized by the formation and collapse of vapor bubbles due to local pressure drops, significantly impacts the hydrodynamic performance and structural stability of high-speed underwater moving bodies [1-2]. Accurate prediction of cavitation features, such as the peak pressure at corresponding displacement, is essential for design optimization, which would lead to a more stable trajectory and faster moving speed [3]. However, the inherent instability and rapid evolution of cavitation flows pose significant challenges for traditional analysis methods.

1.1 Traditional Methods for High-Speed Underwater Slender Body Cavitation Analysis

Physical tests could most accurately demonstrate the motion of such slender bodies moving underwater. For example, Zhao experimentally investigated the cavitation and ballistic characteristics of an asymmetric slender body underwater using high-speed photography. The experiment examined cavity formation, development, collapse, and their impact on the body's trajectory and stability^[4-5]. Since the proposal of Logvinovich theorem, there has also been studies to formulate the cavitation process as well^[6-8]. Complementing experiments, numerical simulations based on Computational Fluid Dynamics (CFD) have become indispensable. Researches on Reynolds-Averaged Navier-Stokes (RANS) equations^[9-10], Large Eddy Simulation (LES)^[11], or multi-phase models like Volume of Fluid (VOF)^[12-13] have successfully captured various aspects of cavitating flows. However, high-fidelity CFD simulations for cavitation analysis, such as those employing multiphase models and turbulence modeling, require extremely fine spatial discretization and small time steps to resolve transient cavity dynamics accurately. A single simulation for an underwater moving body can consume days to weeks of computational time on high-performance clusters. Consequently, both experimental and high-fidelity numerical approaches often result in limited data sets, creating a bottleneck for rapid design iterations. Thus, both industry and academia are seeking for a more cost-efficient way such as machine learning models to simulate motion-related features of such underwater slender bodies.

1.2 Data Driven Surrogate Model for Cavity Prediction

The blooming of Artificial Intelligence (AI) offers promising alternatives through data-driven surrogate models, which can learn complex, non-linear relationships from existing data, providing fast and cost-effective predictions. In related fields, researchers have begun exploring these techniques^[14]. For example, Huang used CFD-generated data to train Artificial Neural Networks (ANNs) for analyzing supercavitation^[15], while Kamali utilized Random Forest models optimized with Bayesian methods to predict ventilated supercavitation characteristics^[16-17]. Oneto developed a Deep Neural Network (DNN) incorporating convolutional layers to predict propeller cavitation noise^[18]. A common theme in these studies is the handling of high-dimensional data, often addressed through feature extraction techniques. Wang applied Principal Component Analysis (PCA) for dimensionality reduction and feature extraction in cavitation flow around hydrofoils, demonstrating its utility in uncovering dynamic evolution patterns^[19]. Similarly, Guang proposed a data-driven reduced-order modeling (ROM) method combining neural networks to predict key modes in cavitation flows^[20]. To further address the challenges of non-linearity and parameter interaction in complex cavitation flows, recent studies have employed more advanced machine learning frameworks.

Machine learning approaches mentioned above have shown promise in cavitation research but face notable limitations in small-sample settings. Prior studies such as using ANNs for supercavitation and applying DNNs for propeller noise, reveal that complex models often overfit and generalize poorly when data are scarce, a phenomenon exacerbated by high dimensionality as discussed studies^[21]. Even Bayesian-optimized models struggle to extract meaningful features from limited cavitation pressure data, limiting their accuracy and reliability. These shortcomings highlight an urgent need for techniques that can enhance model robustness under data constraints. One method to mitigate this problem, feature extraction becomes paramount. It serves to distill the most essential information from high-dimensional raw data, reducing redundancy and noise, thereby improving model robustness and interpretability. Different feature extraction methods offer distinct advantages. Linear methods like Principal Component Analysis (PCA) are computationally efficient and provide interpretable components by maximizing variance, but may miss complex nonlinear relationships^[22-23]. Independent Component Analysis (ICA)^[24-25], by

seeking statistically independent components, can uncover underlying source signals, potentially offering more physically meaningful features for complex flows like cavitation. Meanwhile, non-linear methods like Convolutional Auto-Encoders (CAEs) [24-25] can learn hierarchical representations and capture local patterns in sequential data, such as pressure fluctuations over time or displacement.

Our work addresses this gap by systematically evaluating feature extraction methods—PCA, Fast ICA, and Conv1D AE—tailored for small-sample cavitation pressure analysis. By reducing dimensionality and preserving critical patterns, our approach enables efficient reconstruction and prediction without extensive data, offering a practical solution to accelerate design cycles, to reduce manual intervention, and to provide insights for future applications of machine learning in fluid mechanics.

2. Data Generation and Preprocessing

2.1 Numerical Calculation for Data Generation

Numerical calculations for underwater slender body were performed to calculate the motion and its corresponding cavitation process using StarCCM+. The slender-body was depicted in Fig. 1 with its length(L) and diameter(D) denoted by L and D , respectively. Both the background domain and the subdomain around the slender body were partitioned into Cartesian orthogonal grids in Fig. 2(a). Prismatic grids were applied near the surface of the slender body to capture boundary layer effects in Fig. 2(c). Mesh refinement was performed in regions adjacent to the slender body within the subdomain and the motion range of the slender body in the background domain guided by computational accuracy requirements and overset grid interpolation matching criteria, shown in Fig. 2(a) and Fig. 2(b). All boundaries of the background domain were treated as pressure outlet boundaries, where the pressure under the free surface was set to hydro-static pressure at the local water depth. The region above the free surface was set to atmospheric pressure. The computational domain was discretized into approximately 8.5 million structured grid cells. To accurately resolve the near-wall flow and ensure the validity of the wall functions employed by the Realizable $k-\epsilon$ turbulence model, the wall-mounted y^+ values were controlled to remain within the range of 30 to 100. A baseline adaptive time step of 1×10^{-4} seconds was employed, governed by the Courant-Friedrichs-Lewy (CFL) condition to maintain numerical stability for the high-speed (8-12 m/s) transient cavitating flow. Regarding to the physical models, the realizable $k-\epsilon$ turbulence model was selected for its proven robustness and accuracy in handling high-Reynolds number external flows with strong pressure gradients, such as those encountered during cavitation. The phase change process was modeled using the Schnerr-Sauer cavitation model, which is based on the Rayleigh-Plesset equation and is widely validated for engineering cavitation simulations due to its physical basis in bubble nucleus density and radius.

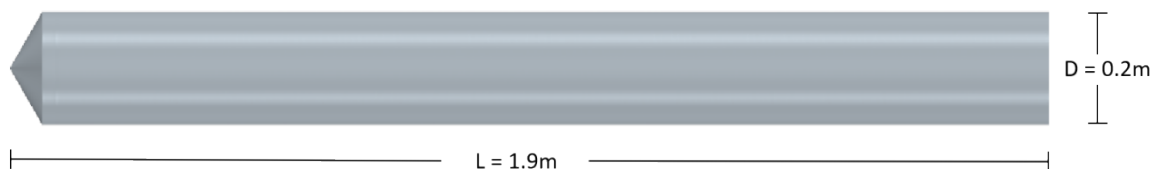


Fig.1 Profile of the slender body

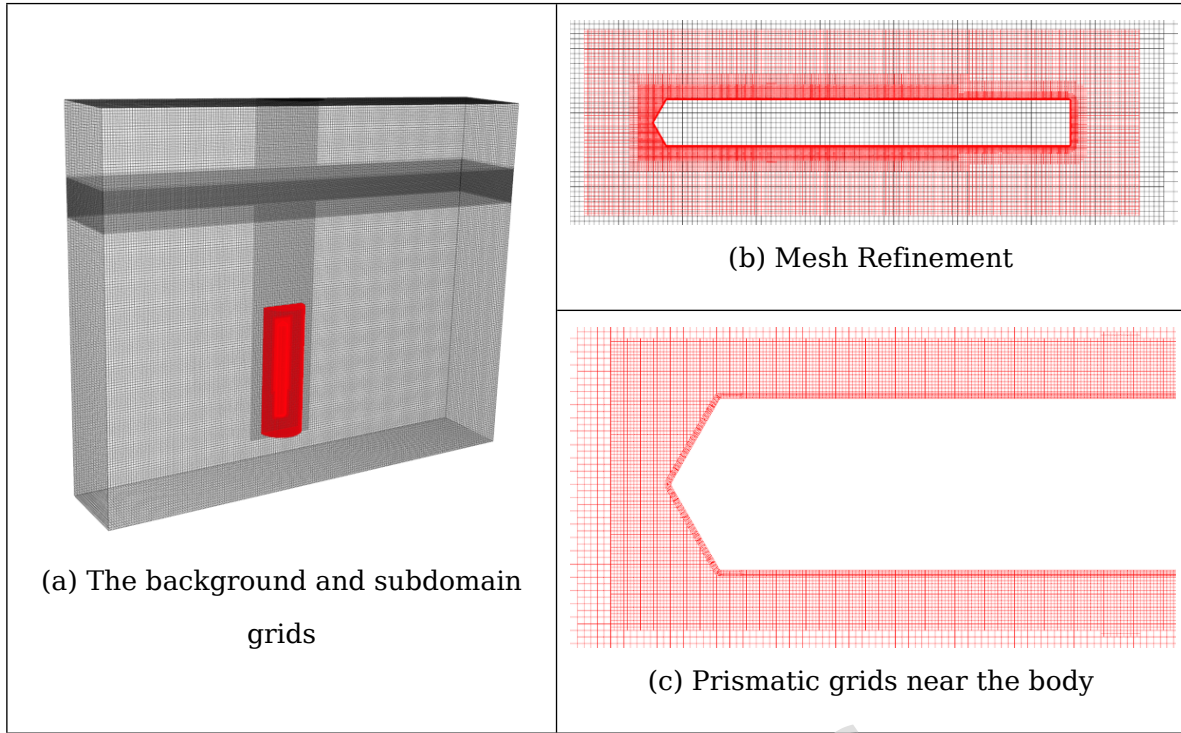


Fig.2 Grids of the simulation Grids of the simulation for a slender body of $L=1.9$ m and $D=0.2$ m

The slender body moved vertically upward, while the background mesh remains stationary. The numerical simulation included the following setup: The vertical motion spanned from an initial depth of 5 meters below the free surface until the body reached the water interface. The velocity was varied between 8 m/s and 12 m/s, corresponding to Reynolds numbers in the range of approximately 1.27×10^7 to 1.9×10^7 , based on the body diameter of 0.2 m. This range encapsulates typical operational conditions for high-speed cavitation scenarios, ensuring the simulations cover relevant hydrodynamic regimes. At the junction between the conical nose and the cylindrical part of the slender body, cavitation occur due to the reduction in pressure. The flow field was simulated by solving the three-dimensional Reynolds-averaged equations with the VOF (Volume of Fluid) model for multi-phase flow, and the control equations were:

$$\frac{\partial \rho_m}{\partial t} + \frac{\partial}{\partial x_j} (\rho_m u_j) = 0 \quad (1)$$

$$\frac{\partial}{\partial t} (\rho_m u_i) + \frac{\partial}{\partial x_j} (\rho_m u_i u_j) = - \frac{\partial P}{\partial x_i} + \rho_m g_i + \frac{\partial}{\partial x_j} [(\mu_m + \mu_T) (\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})] \quad (2)$$

$$\frac{\partial}{\partial t} (\rho_m T) + \frac{\partial}{\partial x_j} (\rho_m u_j T) = - \frac{\partial P}{\partial x_j} + \rho_m g_j + \frac{\partial}{\partial x_j} (\frac{k}{c_p} \frac{\partial T}{\partial x_j}) + S_T \quad (3)$$

Equation (1) represents the conservation of mass for the mixture, where ρ_m is the mixture density of the fluid, t is time, and ∂_j denotes the velocity component in the i -th direction. Equation (2) is the momentum conservation equation, where P is the pressure, ∂_j is the gravitational acceleration component, and μ_T is the turbulent viscosity, which is modeled by the turbulence model to account for the effects of turbulence. Equation (3) describes the energy conservation, with T being the

temperature, k the thermal conductivity, ρc_p the specific heat capacity, and S_T a source term. The turbulence model employed a realizable k- ϵ two-layer model. The phase change mass transport equation proposed by Schnerr and Sauer was used to simulate the cavitation phase change process from liquid water to vapor, and vice versa. The related equations were given below, where R_b denoted the bubble radius, n was the number of bubbles per unit volume of liquid, and α_v represented the vapor volume fraction

$$\frac{d}{dt}(\alpha_v r_b) + \frac{d}{dx}(\alpha_v r_b u) = \dot{m} \quad (4)$$

$$\dot{m} = \frac{r_b r_l}{r} \alpha_v (1 - \alpha_v) \frac{3}{R_b} \sqrt{\frac{2}{3} \frac{P_v - P}{r_l}} \quad (5)$$

$$R_b = \frac{\alpha_v}{1 - \alpha_v} \frac{3}{4\rho n} \frac{1}{\sigma} \quad (6)$$

2.2 Data Preprocessing

There were two sets of data samples used in this study. Each data sample was standardized for better model performance using the equation below, where x represented each feature, μ was the mean value and σ was the standard variance of all features.

$$z = \frac{x - \mu}{\sigma} \quad (7)$$

The first data set was an unsupervised sample set, that the given CFD data was firstly transformed in to a structural data frame by spreading out the measured pressure points over the body surface, shown in Fig. 3(left) below. Each line represented a particular measuring point across the underwater moving body surface. The x-axis represented the displacement of body moved under the water, while the z-axis represented the pressure changes at this point. The y-axis was the channel number, which was a naming routine to distinguish the position of the point. First data set had 559 samples generated from 4 different operating conditions with velocities ranging from 8 to 12 m/s, using the same slender body geometry (length $L=1.9$ m, diameter $D=0.2$ m). The dataset comprises 795 features of pressure points across the underwater slender bodies without labeled outputs. The visualization clearly showed the complexity of the pressure change at different measuring points. The second data set was a supervised sample set, which has 68 samples with 795 features of pressure points across the underwater slender body, and 1 manually labeled outputs of cavity peak pressure. For better clarity, only three samples were visualized in Fig 3(right) with the manually labeled peak pressure the point had. One thing to notice was that the peak pressure one point experienced was different than the maximum pressure during the whole process, which cannot be simply selected from a statistical criterion.

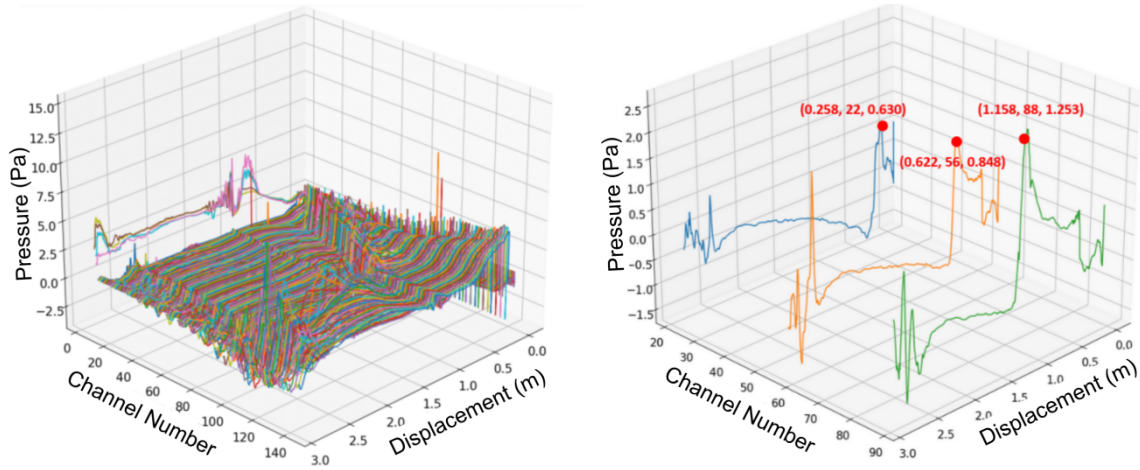


Fig.3 Visualization of unsupervised(left) and supervised(right) data sets

3. Methodologies

This study mainly compared the performance of three different feature extraction methods with two sets of experiments. First experiment applied three feature extraction methods to the unsupervised samples, and tested the ability of these models to reconstruct the original pressure samples. The second experiment constructed a tuned neural network with the pressure across the surface of underwater slender bodies as input, to predict the cavity peak pressure at given measuring points. Given the number of labeled supervised sample was very limited, this experiment applied three different feature extraction methods to analyze the effect on prediction accuracy. The overall process was shown in Fig. 4. below. The detailed models and process were discussed in this section. All machine learning and neural-network based models used in this study were run on a 64G RAM and i7 CPU computer with NPU used for network training acceleration.

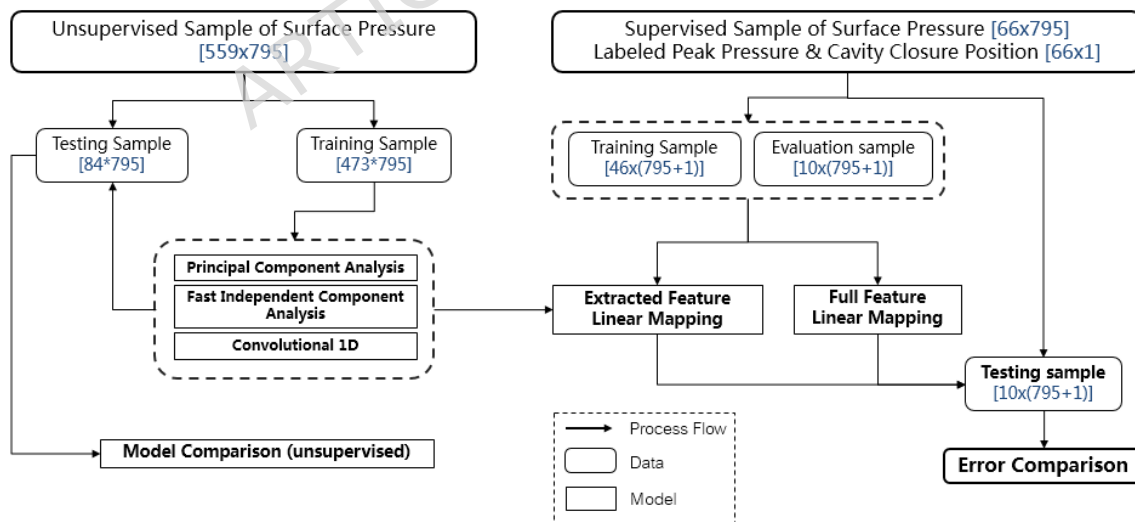


Fig. 4 Flow chart of methodology for overall study process

3.1 Feature Extraction Method

This study selected three different feature extraction models, including Principal Component Analysis, Fast Independent Component Analysis and Auto-Encoder from neural network. This section would describe these methods in detail.

Principal Component Analysis

In the context of cavitation pressure analysis, the pressure fluctuations across a moving body's surface can be viewed as a complex signal composed of multiple underlying physical processes. Principal Component Analysis (PCA) serves as a powerful linear technique to simplify this complexity. Imagine the pressure data as a high-dimensional cloud of points where each dimension represents the pressure at a specific displacement or time. PCA works by identifying the orthogonal directions (principal components) in this cloud that capture the maximum variance [21]. The first principal component aligns with the direction of greatest spread in the data, which often corresponds to the most dominant trend in the pressure evolution, such as the overall pressure drop associated with cavitation inception. Subsequent components capture progressively smaller, orthogonal variations. This is analogous to finding the optimal viewing angles to see the main shape of a complex object, filtering out minor irregularities. By projecting the original high-dimensional pressure data onto a reduced set of these principal components, PCA effectively distills the essential spatial-temporal patterns while reducing noise and redundancy. This makes it particularly suitable for reconstructing the primary structure of the pressure evolution, though its linear nature may limit its ability to capture highly nonlinear interactions inherent in turbulent cavitation flows.

The unsupervised data set could be represented as a matrix of $X \in \mathbb{R}^{n \times d}$, where n is the number of samples and d is the number of features. The key steps of PCA can be summarized as follows. The first step was to center the data by subtracting the mean of each feature, denoted as $\tilde{X} = X - 1\mu^T$, where μ is the mean vector of the features. Then the covariance matrix C was computed as: $C = \frac{1}{n-1}\tilde{X}^T\tilde{X}$. The covariance matrix C was then decomposed into its eigenvalues and eigenvectors. The eigenvalues λ_i and corresponding eigenvectors v_i satisfied the equation $Cv_i = \lambda_i v_i$. The eigenvectors were sorted in descending order of their corresponding eigenvalues, which represented the amount of variance explained by each principal component. To reduce the dimension, top k eigenvectors corresponding to the largest k eigenvalues were selected. These eigenvectors formed the columns of the transformation matrix $P \in \mathbb{R}^{d \times k}$. The reduced-dimension data $Y \in \mathbb{R}^{n \times k}$ is then obtained by $Y = \tilde{X}P$.

PCA had several advantages, including its simplicity, interpretation, and ability to handle high-dimensional data. However, it assumed linear relationships between variables and may not capture nonlinear patterns. This could be a drawback of this method since the pressure points on a single longitude could have nonlinear relationships. Visualization of the PCA model results of the unsupervised sample was demonstrated in Fig. 5 (left). Each axis represented one feature obtained from this model.

Fast Independent Component Analysis

While PCA seeks components that are orthogonal to each other, Fast Independent Component Analysis (Fast ICA) aims to uncover statistically independent source signals that mix together to form the observed pressure data [22][23]. Physically, the measured pressure signal at any point on the body surface can be thought of as a mixture of contributions from various independent physical phenomena, such as the potential flow around the body, bubble collapse events, vortex shedding, and structural vibrations. Fast ICA operates on the principle that these underlying sources are statistically independent and have non-Gaussian distributions. The algorithm attempts to reverse the mixing process, much like separating individual voices from a recorded conversation in a noisy room. It finds a linear transformation that maximizes the statistical independence of the components, often by maximizing their non-Gaussianity. For cavitation pressure signals, this means Fast ICA might isolate

specific, physically meaningful source signatures—like a distinct pattern associated solely with a cavity collapse event—from the composite measured signal. This capability to unmix independent sources makes Fast ICA a promising tool for identifying and analyzing distinct physical mechanisms within the complex cavitation process, potentially offering deeper insight than variance-based methods like PCA.

Mathematically, the sample data X was expressed as $X = AS$, where A was the mixing matrix, and S represented the independent source components. The task of Fast ICA was to estimate the un-mixing matrix W , such that $S = WX$, where the components of S were as statistically independent as possible. The core of Fast ICA was its optimization framework. The algorithm aimed to maximize the non-Gaussianity of the latent representations, as independence was closely tied to non-Gaussianity in the context of ICA. The optimization was performed using a fixed-point iteration scheme, which iteratively updates the un-mixing matrix W to maximize the non-Gaussianity of the outputs. The update rule for W was derived from the contrast function, which measured the deviation from Gaussianity. A commonly used contrast function was the negative-entropy, defined as:

$$J(y) = E\{g(y)\} - E\{\log E\{g'(y)\}\}$$

where $g(y)$ was a non-quadratic function, such as $\cosh(ay)$ or $\exp(-ay^2)$, and a was a constant that controls the curvature of the function. The gradient of the contrast function with respect to W drove the optimization process. The gradient update rule for W was:

$$W_{k+1} = W_k + \mu \cdot \text{diag}(E\{g'(W_k^T X)\}) - E\{g''(W_k^T X)\} \cdot W_k$$

where μ was the learning rate, and the expectation was taken over the data distribution. This iterative process continued until W converges to a stable solution.

The key advantage of Fast ICA was its ability to handle high-dimensional data efficiently. The algorithm's fixed-point iteration scheme ensured rapid convergence, making it computationally feasible for large-scale datasets. Additionally, the use of non-Gaussianity as an independence criterion allowed Fast ICA to uncover latent structures that are often missed by linear methods like PCA, which rely solely on variance maximization. Visualization of the Fast ICA model results of the unsupervised sample was demonstrated in Fig.5 (right). Each axis represented one feature obtained from this model.

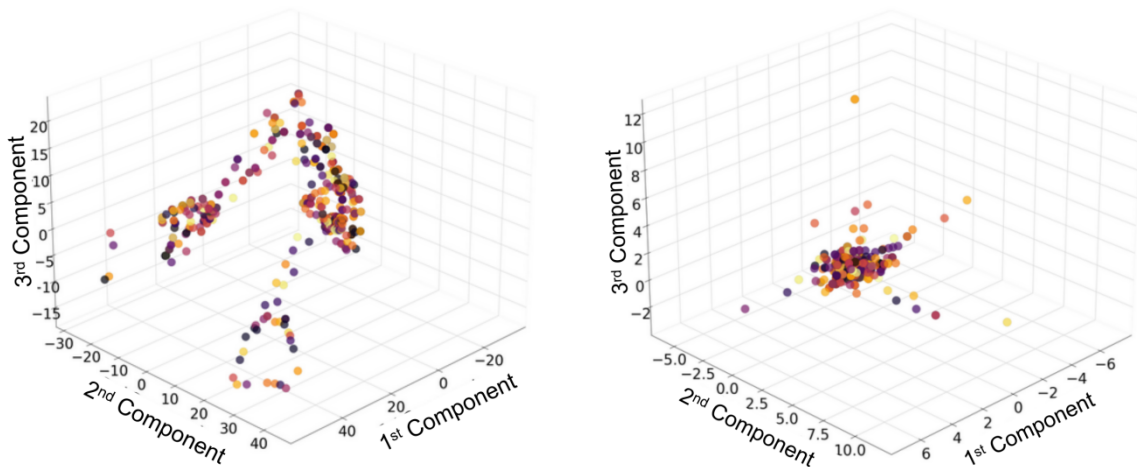


Fig. 5 Top three features extracted by PCA (left) and Fast ICA (right)

Convolutional 1D Auto-Encoder

Convolutional 1D Auto-Encoders (Conv1D AEs) belong to a class of non-linear neural networks designed for efficient feature learning from sequential data, such as the pressure evolution over displacement [24-25]. The core idea was to use a neural network to learn a compressed, latent representation of the input data in an unsupervised manner, as shown in Fig. 6 below. The network consists of two parts: an encoder and a decoder. The encoder processed the input pressure sequence using one-dimensional convolutional filters that slide along the displacement axis. These filters act as feature detectors, learning to recognize local patterns—such as a sharp pressure peak or a gradual recovery—at different scales. Through operations like convolution and pooling, the encoder compressed the input into a compact latent-space representation. The kernel and padding size of each convolution layer, and maximum pooling window size were shown in Figure 6, and each layer was connected by a ReLu activation layer. The latent size was a 32x4 tensor as shown. The decoder then reconstructed the original pressure sequence from this latent representation. Similarly, the kernel and padding size of each convolution layer, and upsampling size were shown in Figure 6, and each layer was connected by a ReLu activation layer. The model was trained by minimizing the difference between the original and reconstructed pressure signals. Unlike the linear transformations of PCA and Fast ICA, the Conv1D AE could capture complex, non-linear relationships in the data. This made it potentially powerful for learning hierarchical features from cavitation pressure signals.

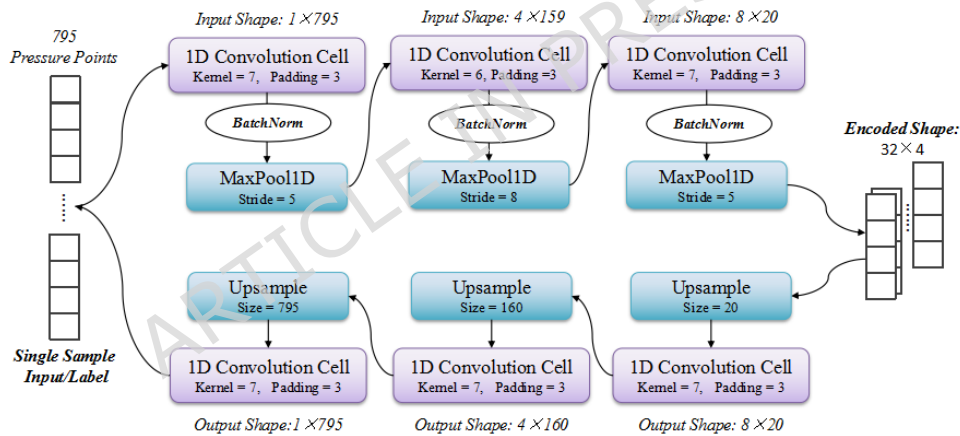


Fig. 6 Structure of the customized Conv1D Auto-encoder network

The encoder's output was defined as $z = f_e(x; \theta_e)$ where f_e was a nonlinear activation function, and θ_e represented the encoder's parameters. Conversely, the decoder's output is $x' = f_d(z; \theta_d)$, with θ_d as the decoder's parameters. The model minimized the reconstruction loss, typically using mean squared error: $L = \|x - x'\|^2$, or cross-entropy for probabilistic outputs. In comparison to other dimension reduction techniques, such as PCA, the 1D CAE offers a nonlinear transformation, enabling the discovery of more complex and abstract features inherent in the data. This nonlinear nature allowed the model to uncover intricate patterns that linear methods might overlook.

3.2 Linear Mapping Model

To approximate the peak pressure and closure position, a direct mapping method of a neural network consisted of linearly connected neurons was constructed. The linear mapping model, which was denoted as 'Full Feature Mapping Model' in Fig.7, took all 795 pressure value on a single measuring point across the slender body's

surface as inputs, and through the activation layers to approximation the peak pressure. The 'Extracted Feature Linear Mapping' had the same setting except for the different input of extracted features. Layers with 32-16-8 neurons were set up for all trials. Activation layer of Relu were applied after each layer. To reduce the chance of model over-fitting, a drop-out rate of 0.1 were set. The 0.001 learning rate and Adam optimizer were selected via grid search for stable convergence. 800 Training epochs were determined using early stopping to prevent overfitting, consistent with common practice in small-sample learning.

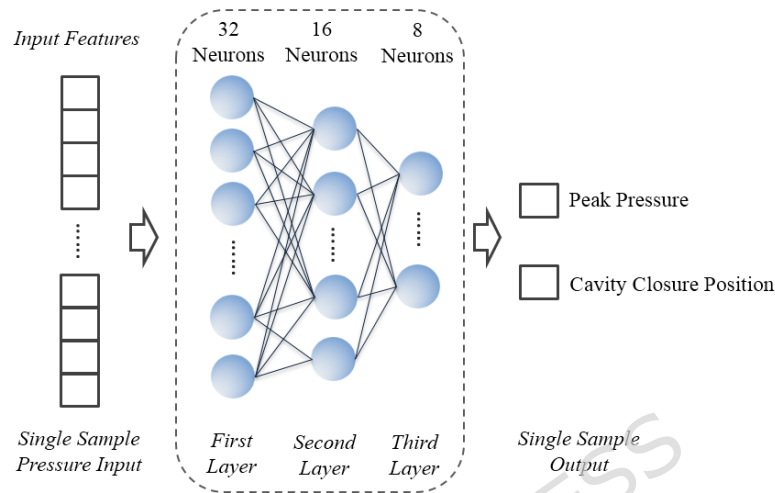


Fig. 7 Structure of Linear Mapping Model

4. Unsupervised comparison of Feature Extraction Performance

According to the overall methodology, three feature extraction methods were firstly applied to the training part of unsupervised samples. The trained models were used on the testing samples to evaluate the ability of reconstruction from the extracted features. Each model was trained to represent at least 99% of the original sample information in terms of variance explained or reconstruction error depending on the type of methods used. The trained models were then applied on three individual testing sample. The reconstruction errors were compared to evaluate the model performance.

Number of principals from 1 to 100 were used for PCA method on the training data to extract features, and their corresponding variance explained were compared, shown in Fig. 8 below. Some of the key components were listed in Table 1 below for better comparison, which showed with 35 components the model was able to represent over 99% variance of the original sample.

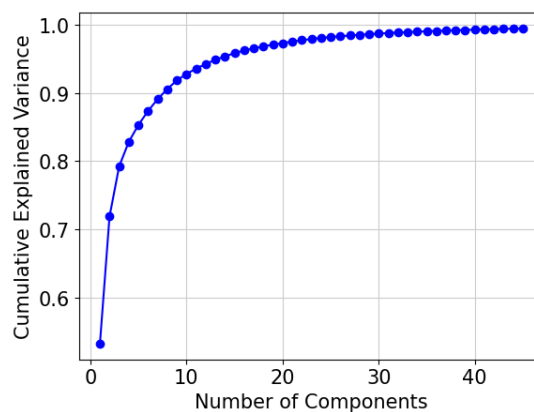


Fig.8 Number of components and the corresponding variance explained of the PCA model

Table 1 Selected components and the corresponding variance explained of the PCA model

No. Component	1	3	5	7	13	23	35	45
Variance Explained	0.719	0.828	0.874	0.906	0.954	0.981	0.990	0.995

Three testing sample were randomly selected to test the reconstruction ability of reconstruction. The reconstructed pressure change over the displacement were shown in Fig.9. From the figure, trend of pressure change could be reconstructed successfully, while there were some under-estimation of peak value and some over-fitted small pressure changes.

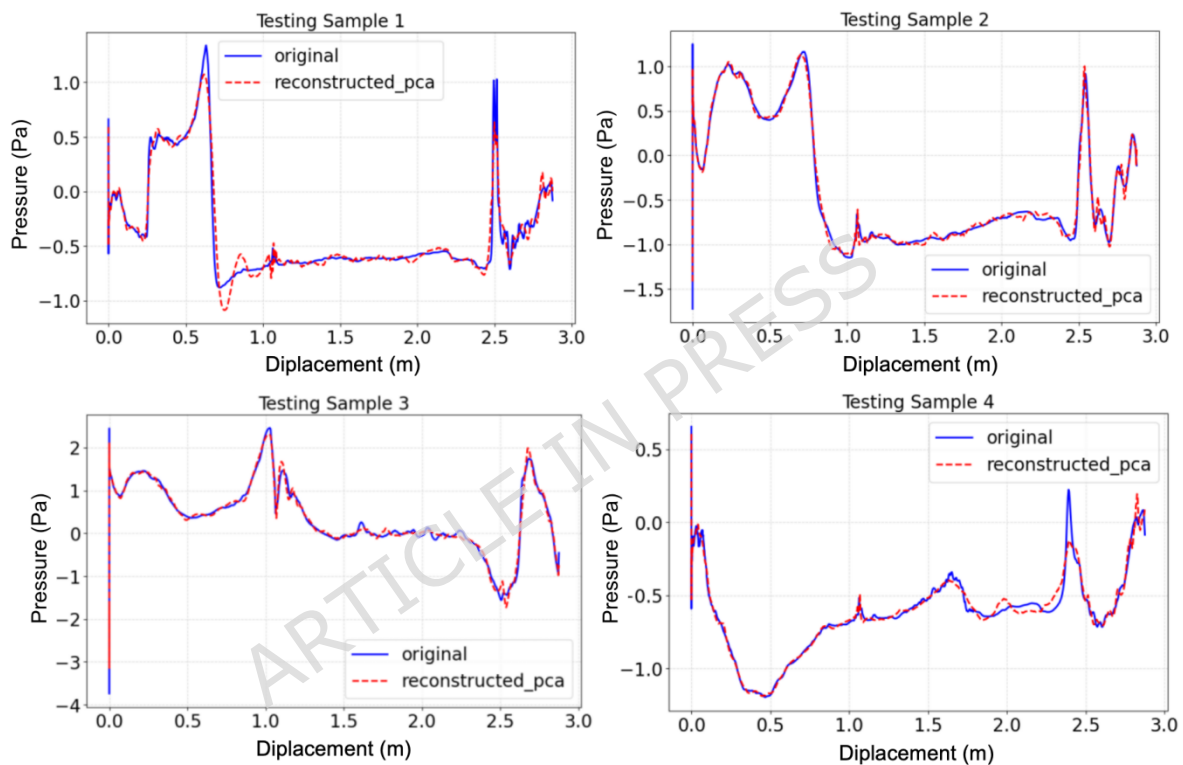


Fig.9 Reconstructed testing samples by PCA feature extraction model

1-100 components were trained on the training sample by Fast ICA method, which focused on seeking the independent component instead of maximizing the variance explained. The reconstruction error of different number of components was shown in Fig.10 and Table 2 below. Similarly, 35 components could represent at least 99% of the original data information.

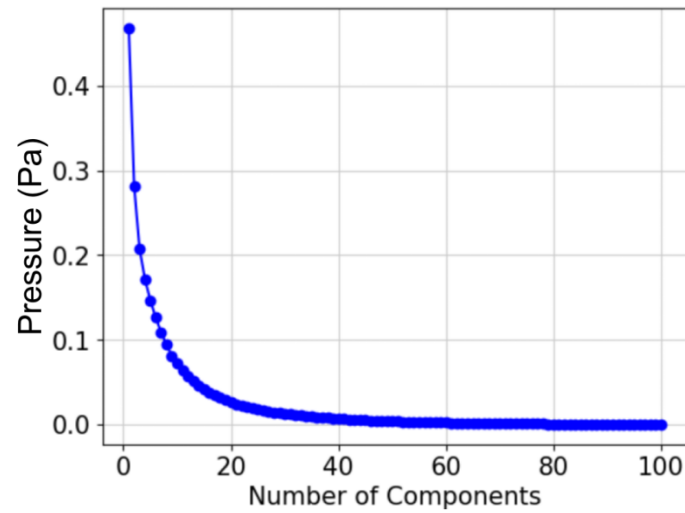


Fig.10 Number of components and the corresponding reconstruction error of the Fast ICA model

Table 2 Selected components and the corresponding reconstruction error of the Fast ICA model

No. Component	1	3	5	8	14	24	35	47
Variance Explained	0.468	0.208	0.147	0.094	0.046	0.019	0.010	0.005

The same three testing sample were randomly selected to test the reconstruction ability of reconstruction. The reconstructed pressure change over the displacement were shown in Fig.11. From the figure, the prediction performance was close to that of PCA.

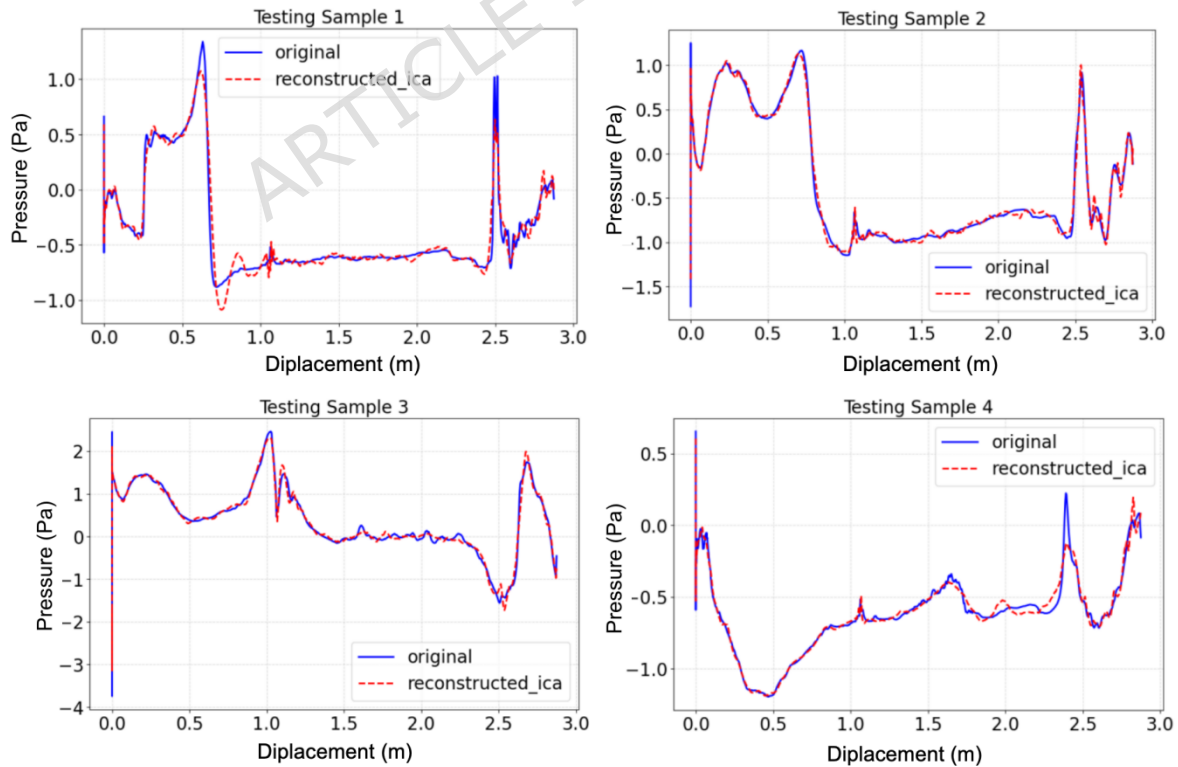


Fig.11 Reconstructed testing samples by Fast ICA feature extraction model

The convolution 1D network described earlier were used to train the model. A batch size of 795 was used, matching the total number of samples to utilize the full dataset

for each gradient update. The Adam optimizer parameters were set to their typical default values ($\beta_1=0.9$, $\beta_2=0.999$), which are widely recognized for providing stable convergence in deep learning applications. This configuration was selected to ensure efficient and consistent model training. A consistent learning rate of 0.1 was applied through the whole training process. The epoch loss for 1000 epochs was shown in Fig.12. The converged loss was around 1.0%. The model performance on three tested samples was shown in Fig.13. In contrast to the occasional over-fitting phenomenon of PCA and Fast ICA, Conv1D model demonstrated under-fitting in sample 3 and 4, which meant the peaks were ignored by the model.

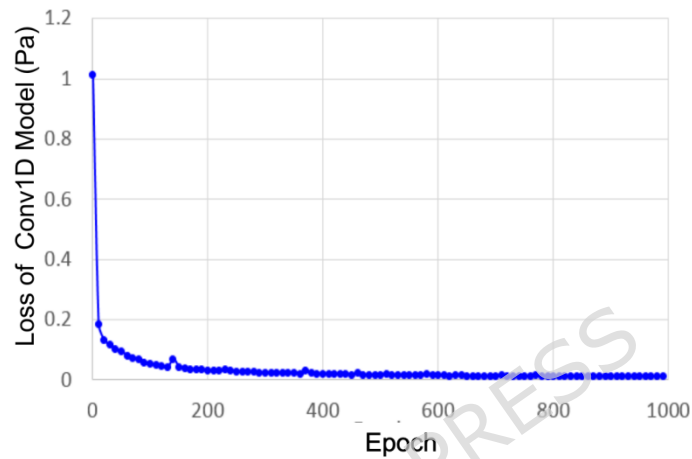


Fig.12 Epoch loss of training Conv1D Auto-encoder

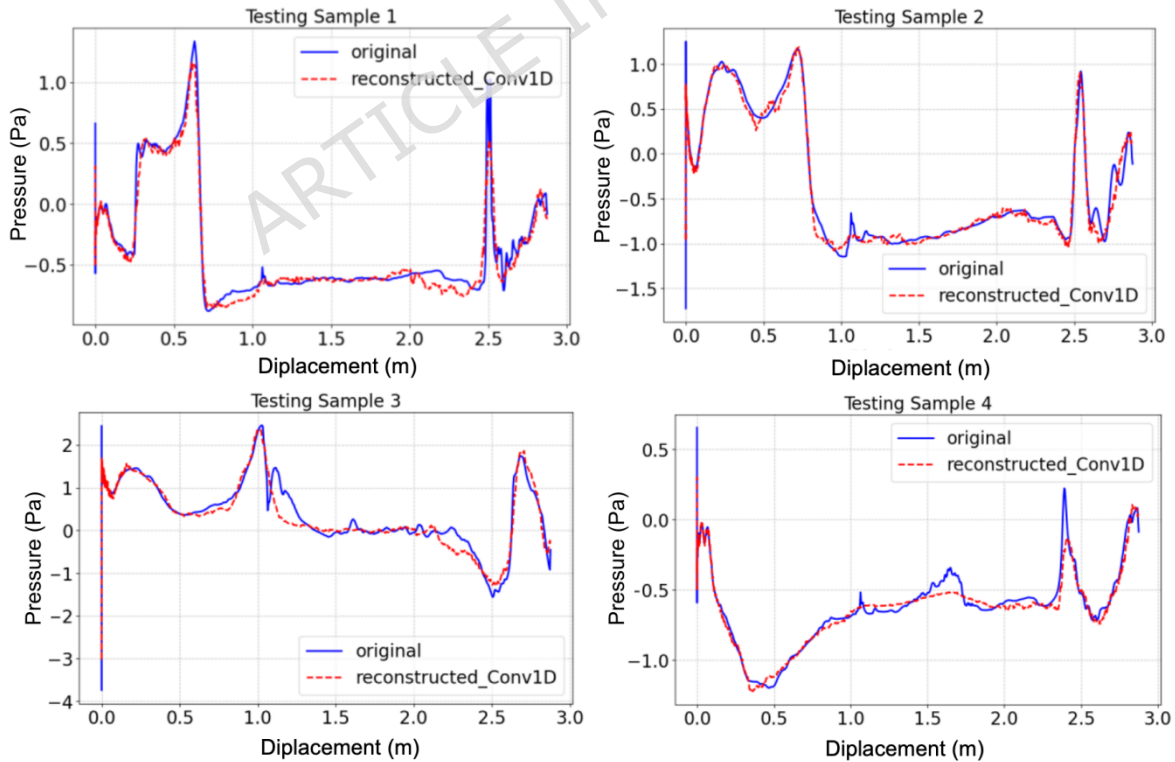


Fig.13 Reconstructed testing samples by Conv1D auto-encoder feature extraction model

The observed under-fitting of sharp pressure gradients by the Conv1D Auto-encoder could be attributed to its architectural properties. The Maxpooling layers within the

encoder, while effective for dimensionality reduction, can inadvertently suppress high-frequency spatial signals corresponding to rapid cavitation events like local bubble collapses. This inherent smoothing effect contrasts with the proficiency of linear methods.

All testing samples were tested by three models, and the average reconstruction error were summarized in the table below. Classic statistical criteria of Mean Square Error (MSE), Mean Absolute Percentage Error(MAPE) and Coefficient of Determination (R^2). Fast ICA had the best performance in all criteria in terms of the ability to reconstruct the original pressure change with extracted features. The observed under-fitting by the Conv1D AE in the displacement range of 1.0-2.5, where it fails to capture sharp pressure gradients, can be attributed to its architecture. The max-pooling layers within the encoder, while effective for dimensionality reduction, may excessively smooth localized, high-frequency features crucial for representing rapid cavitation dynamics. This highlights a limitation of the current network design in preserving critical transient information compared to methods like PCA and Fast ICA.

Table 3 Reconstruction Error of all three models on testing samples

Model	PCA	Fast ICA	Conv1D AE
MSE	0.012	0.010	0.018
MAPE	0.365	0.350	0.578
R^2	0.987	0.989	0.982

5. Supervised comparison of Feature Extraction Performance

With the three trained feature extraction models, experiments were carried out to evaluate the effectiveness of these extracted features on predicting the peak pressure. A linear mapping models described in Section 3.2 was constructed and trained given the pressure change of a particular measuring point on underwater moving bodies. To minimize the effect of training process, the mapping models for all experiment trials were set to the exact same network structure with a constant learning rate of 0.001, an Adam optimizer, MSE loss and a batch size of 16. Each training process lasted for 800 epochs. The training loss and evaluation loss of all trials were shown in Fig.14 below.

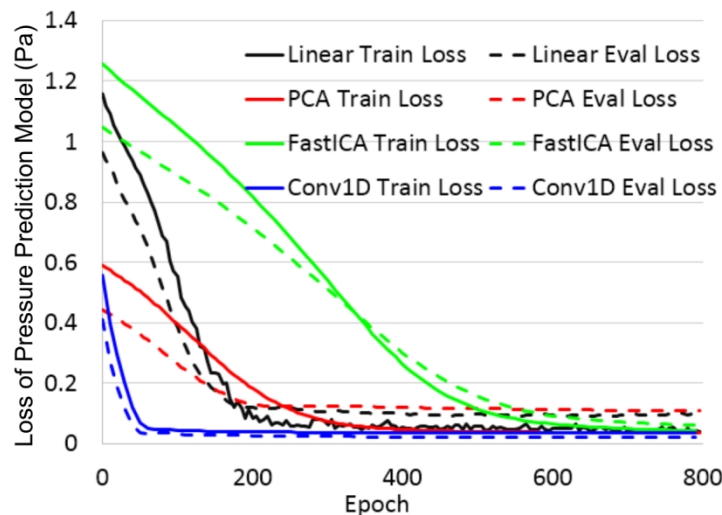


Fig.14 Loss comparison of peak pressure prediction models

From the figure, all evaluation loss and training loss converged at the end of the total training process, which implied no significant over-fitting and under-fitting situations. The black solid and dashed lines presented the input of original sample without feature extraction methods applied. The blue lines represented samples treated by Conv1D Auto-encoder model, which converged faster than the original sample. This could mean the extracted features Conv1D Auto-encoder had a better mapping ability to the peak pressure. However, the Fast ICA models had a very different performance than PCA although they had similar performance on reconstructing the unsupervised samples.

The performance of the models, evaluated using 5-fold cross-validation, is summarized in Table 4. The values represent the mean and standard deviation of the Mean Absolute Percentage Error (MAPE) across all validation folds. For illustrative purposes, the mean prediction results for are visualized in Fig. 15. 'Direct' meant the original sample was directly put into the mapping model without feature extraction models added. From the figure and numbers, PCA and Conv1D Auto-encoders both improved the prediction accuracy compare to the original sample by comparing the mean absolute percentage error. Conv1D reduced the prediction error by 10%, which could be helpful in real-world engineering scenarios.

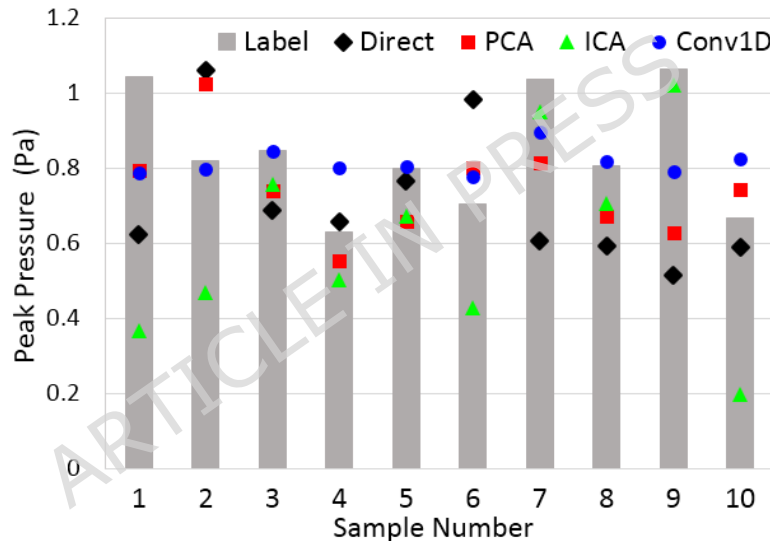


Fig.15 Peak pressure prediction result

Table 4 Peak pressure prediction performance

Model	Direct	PCA	Fast ICA	Conv1D AE
MAPE(%)	32.5 ± 3.1	29.2 ± 2.8	41.8 ± 4.5	22.3 ± 2.0

The divergent performance of Fast ICA stem from a fundamental conflict between its mathematical assumption of statistical independence and the underlying physics of the peak pressure event. The peak pressure was not an independent source signal; but a highly nonlinear culmination of interdependent physical processes. Fast ICA excelled at reconstruction by decomposing the pressure signal into statistically independent components, effectively separating the underlying physical phenomena. However, the peak pressure was not an independent event. It was a highly nonlinear, transient culmination of interdependent processes like cavity collapse and vortex dynamics, which violated ICA's core assumption of statistical independence. In contrast, the Conv1D Auto-Encoder imposed no such independence constraint. It learnt localized, hierarchical features directly from the data, allowing it to capture the specific nonlinear signatures that led to the peak pressure, making its features more predictive for this specific, complex task.

6. Conclusions

This study aimed to investigate the effect of three feature extraction models on the cavitation-related feature of underwater moving bodies given a small sample size. Model performance was evaluated by both reconstructing the pressure change in an unsupervised setting and predicting the peak pressure across the moving bodies' surface with limited supervised samples. The results demonstrated that Fast ICA had the best reconstruction ability of pressure change against displacement of moving bodies, followed by PCA and Conv1D Auto-encoder. All methods could have a reconstruction MSE less than 0.02. However, when the extracted feature patterns were used on the limited supervised samples to predict the peak pressure of each measuring points, Conv1D Auto-encoder showed the best performance in reducing the MAPE by 10% compared to direct prediction model without feature extraction followed by PCA, while Fast ICA increased the error. These findings firstly proved that appropriate feature extraction methods could be effective in learning the underlying patterns of pressure change for underwater moving bodies. This could reduce the reliance of expert knowledge. Also, applying the extracted-feature prediction model for the peak pressure could reduce the manual work of locating the pressure points, thus expedite the analysis for hydrodynamic performance of such bodies in engineering scenarios. This work extends prior applications of PCA in cavitation analysis by systematically comparing its efficacy against Fast ICA and Conv1D AE specifically for small-sample scenarios. Unlike studies focusing on propeller cavitation noise or ventilated supercavitation, our approach targeted peak pressure prediction on moving bodies, demonstrating Conv1D AE's superior accuracy where traditional linear methods like ICA underperform.

While the proposed feature extraction methods, particularly the Conv1D AE, showed promising results in peak pressure prediction under a small-sample setting, this study had several limitations. The primary limitation was the constrained size of the dataset, which, despite the use of cross-validation, ultimately limited the complexity of the models that can be reliably trained and the generalizability of the findings to vastly different flow conditions or body geometries. The performance hierarchy observed may be specific to the current data distribution. While results demonstrate effectiveness within our dataset, robustness against more varied and highly nonlinear cavitation regimes requires further investigation. Moreover, The CFD model assumed specific turbulence and cavitation models, which simplified the complex physics.

The study was a preliminary work on finding effective feature extraction method. Future work will focus on expanding the dataset to include a wider range of operating conditions and body shapes to further validate and enhance the robustness of the models. Additionally, exploring other regularization techniques specifically designed for small-sample learning with deep neural networks, such as specific data augmentation strategies tailored for fluid dynamics data, is a promising direction. With the advance in machine learning technique, there is great potential of implementing such tools to accelerate analyzing and designing process for underwater moving bodies.

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